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The dominating 4-colour theorem

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A *dominating K_t-model* in a graph G is a sequence $(T_1, \dots, T_t)$ of pairwise vertex-disjoint connected subgraphs of G, such that whenever $1 \leqslant i < j \leqslant t$ every vertex in T_j has a neighbour in T_i. Replacing "every vertex in T_j" by "some vertex in T_j" retrieves the standard definition of K_t-model, which is equivalent to a K_t-minor in G. We prove that every graph with no dominating K_5-model is 4-colourable. This generalises and is significantly stronger than the 4-colour theorem for planar graphs or for graphs with no K_5-minor. It also makes progress towards Hajós' conjecture on K_5-subdivisions in 5-chromatic graphs.

1 Introduction

The 4-colour theorem states that every planar graph is 4-colourable [2, 35]. By the Kuratowski–Wagner theorem [31, 38], the 4-colour theorem is equivalent to saying that every K_5-minor-free and $K_{3,3}$-minor-free graph is 4-colourable. In fact, using this and Wagner's characterisation of K_5-minor-free graphs, it is straightforward to show that every K_5-minor-free graph is 4-colourable.

We prove a substantial generalisation of the latter result via the following definition introduced by [26]. A *dominating K_t-model* in a graph G is a sequence $(T_1, \dots, T_t)$ of pairwise disjoint connected subgraphs of G, such that whenever $1 \leqslant i < j \leqslant t$ every vertex in T_j has a neighbour in T_i. Replacing "every vertex in

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T_j" by "some vertex in T_j" retrieves the standard definition of K_t-model, which is equivalent to a K_t-minor in G. The main result of our paper is the following.

Theorem 1.1. Every graph with no dominating K_5-model is 4-colourable.

The proof of Theorem 1.1 (which employs the 4-colour theorem for planar graphs) is presented in Section 3. Before that we make several comments about Theorem 1.1.

Restricted K_5-Models. Note that Theorem 1.1 is equivalent to saying that every 5-chromatic graph has a dominating K_5-model $(T_1, \dots, T_5)$. If w is any vertex in T_5 and v is any neighbour of w in T_4, then $(T_1, T_2, T_3, \{v\}, \{w\})$ is a dominating K_5-model [26]. Thus, there is a dominating K_5-model with two singleton branch-sets. This strengthens a result of [32, Cor. 1.4] who showed that every 5-chromatic graph contains a (not necessarily dominating) K_5-model with at least one singleton branch-set. Two singleton branch-sets is optimal here, since there are 5-chromatic graphs with arbitrarily large girth [18], and no complete graph model in any such graph has three branch-sets of bounded size. One final remark on the structure of branch-sets: we may further take T_3 to be a path and $G[V(T_3) \cup V(T_4) \cup V(T_5)]$ to be an induced cycle. (Take a dominating K_5-model in which T_4 and T_5 are singleton branch-sets, $T_4 = \{v\}$ and $T_5 = \{w\}$, say. T_3 must contain a neighbour of v and a neighbour of w and so there is a cycle in $G[V(T_3) \cup \{v\} \cup \{w\}]$ that contains the edge vw. Let C be a shortest cycle in $G[V(T_3) \cup \{v\} \cup \{w\}]$ that contains the edge vw. The minimality of C implies that it has no chords and so is an induced cycle in G. Then $(T_1, T_2, C - \{v, w\}, \{v\}, \{w\})$ is the desired dominating K_5-model.)

Comparison with K_5-Minors. The class of graphs with no dominating K_5-model is significantly more general than the class of graphs with no K_5-minor. For example, the 1-subdivision of K_n has no dominating K_4-model [26], but obviously contains K_n as a minor. Moreover, no graph with maximum degree 3, including random cubic graphs and cubic expanders, has a dominating K_5-model. In fact, every graph, in which no two vertices with degree at least 4 are adjacent, has no dominating K_5-model (since if $(T_1, T_2, T_3, \{v\}, \{w\})$ is a dominating K_5-model, then v and w are adjacent vertices both with degree at least 4). There are even 4-connected graphs that contain a K_5-minor but contain no dominating K_5-model. For example, let G be obtained from the complete bipartite graph $K_{5,5}$ by removing the edges of a perfect matching. Note that G has a perfect matching whose contraction gives a K_5-minor. However, it does not contain a dominating K_5-model. (Let $\{A, B\}$ be the bipartition of the $K_{5,5}$ and M be the matching removed. Suppose towards a contradiction that G has a dominating K_5-model $(T_1, \dots, T_5)$. From the above discussion, we may assume that $T_4 = \{p\}$ and $T_5 = \{q\}$ where $p \in A$ and $q \in B$. Since G is triangle-free, each of T_1, T_2, T_3 has at least two vertices. Suppose that $|T_i| = |T_j| = 2$ for distinct $i, j \in \{1, 2, 3\}$. Let $T_i = ab$ and $T_j = cd$ with $a, c \in A$ and $b, d \in B$. Then $\{a, c, p\}, \{b, d, q\}$ is a bipartition of $K_{3,3}$, which does not exist in G. So at most one of T_1, T_2, T_3 has exactly two vertices. Since $|V(G)| = 10$, two of T_1, T_2, T_3 have exactly three vertices, and the other has exactly two vertices. In particular, every vertex of G is used by $T_1, \dots, T_5$ and T_1 is a path with two or three vertices. Let $v \in V(T_1)$ be the only vertex of T_1 in its side of the bipartition. Then the vertex matched with v in M is not in T_1 and not dominated by T_1, which is the required contradiction.)

Algorithmic aspects. It is routine to turn our proof into a polynomial-time algorithm that given as input a graph without a dominating K_5-model, returns a 4-colouring of that graph. Roughly speaking, the algorithm (whose correctness is supported by the proof) works as follows: As long as the current graph G is not 4-connected, the algorithm splits the graph along a lower-order separation into two smaller graphs G_1, G_2 that still do not have a dominating K_5-model. (Formally, this is done in the setting when we only exclude so-called L-compatible dominating K_5-models, but we skip this technicality for this sketch.) These graphs can then be 4-coloured by applying the algorithm recursively to G_1 and G_2 and combining colourings appropriately (G_1 and G_2 will be defined so that the colourings are indeed compatible at the separator). Here we use that a 4-colouring of an n-vertex planar graph can be computed in $O(n^2)$ time [35]. If G is 4-connected, then our proof shows that contracting a suitable subset of the second-neighbourhood

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of some vertex yields a smaller instance G', which can be recursively 4-coloured and whose colouring can be extended to the full graph G upon uncontracting.

Hadwiger's Conjecture. We now place Theorem 1.1 in the context of some more general conjectures and related problems studied in the literature. Hadwiger [21] famously conjectured that every graph with no K_t-minor is $(t-1)$-colourable. This conjecture is straightforward for $t \leqslant 4$ [15, 21]. It is true for $t = 5$ as mentioned above, it is true for $t = 6$ as proved by [34], and it is open for $t \geqslant 7$. The dominating Hadwiger conjecture, stated by [26], asserts that every graph with no dominating K_t-model is $(t-1)$-colourable. This conjecture implies and is significantly stronger than Hadwiger's conjecture. Illingworth and Wood [26] proved the dominating Hadwiger conjecture for $t \leqslant 4$. Theorem 1.1 proves it for $t = 5$.

In every colouring $c\colon V(G) \to \{1,\dots,t\}$ of a graph with chromatic number t, every pair of colour classes is adjacent. Hence, one intuition for Hadwiger's conjecture is that, if $\chi(G) \geqslant t$, then G contains t disjoint vertex-subsets that are pairwise adjacent (this is a K_t-model without the connectedness condition). Further, if the colouring c is chosen to minimise $\sum_{v\in V(G)} c(v)$, then every vertex in colour class i has a neighbour in each smaller colour class. Hence, the colour classes form a dominating K_t-model again without the connectedness condition. This gives an analogous intuition for the dominating Hadwiger conjecture.

Hajós' Conjecture. Hajós' conjecture is a famous strengthening of Hadwiger's conjecture stating that, for every $t \in \mathbb{N}$, every graph G with $\chi(G) \geqslant t$ contains a *subdivision of* K_t as a subgraph. For $t \leqslant 4$, this is equivalent to Hadwiger's conjecture, since for $t \leqslant 4$ a graph contains K_t as a minor if and only if it contains a subdivision of K_t. However, starting from $t = 5$, Hajós' conjecture becomes significantly stronger than Hadwiger's conjecture, since the equivalence between minor and subdivision containment breaks for $t \geqslant 5$. Catlin [6] famously showed that Hajós' conjecture is false for every $t \geqslant 7$; also see [37]. Indeed, while Hadwiger's conjecture and the dominating Hadwiger conjecture are true for almost every graph [3, 26], Hajós' conjecture is false for almost every graph [19]. Hajós' conjecture remains open in the cases $t \in \{5, 6\}$. In particular, the case $t = 5$ has attracted a significant amount of attention [36, 39–41] and is closely connected to the now-resolved longstanding Kelmans–Seymour conjecture [22–25]. These works combine to show that any minimal counterexample to Hajós' conjecture for $t = 5$ is 4-connected but not 5-connected.

To demonstrate the significant difference between Hadwiger's conjecture and Hajós' conjecture in the case $t = 5$, note that there is an explicit family of 22 non-isomorphic graphs (including the Petersen graph, H_{22}) $H_1,\dots,H_{22}$, depicted in Figure 1, such that a graph G contains K_5 as a minor if and only if it contains a subdivision of at least one of $H_1,\dots,H_{22}$. (Reference [33, (2.1)–(2.3)] showed that for every graph H there is a finite set S of graphs such that a graph G contains H as a minor if and only if G contains a subdivision of some graph in S. We implemented a naïve algorithm in SageMath to compute S when $H = K_5$; the code is available at https://gist.github.com/junghonyip-math/d20dd0703a86d083f24a8ba5450a3765.) While Hadwiger's conjecture for $t = 5$ only guarantees a subdivision of one of these 22 graphs in any 5-chromatic graph, Hajós' conjecture makes the much stronger claim that, out of these 22 graphs, only one is necessary, namely K_5 itself.

Theorem 1.1 quickly implies the following result, which shows that the list of graphs necessary in this statement can be reduced from the list of 22 graphs given by Hadwiger's conjecture to only two graphs: $K_5 = H_1$ and the unique 6-vertex graph $\widehat{K}_5 = H_2$ obtained from K_5 by splitting one vertex into two adjacent degree-3 vertices.

Corollary 1.2. Every graph G with $\chi(G) \geqslant 5$ contains a subdivision of K_5 or $\widehat{K}_5$.

Proof. By Theorem 1.1, there exists a dominating K_5-model $(T_1,\dots,T_5)$ in G. Clearly, the subgraph of G induced by the vertices in $V(T_2) \cup \dots \cup V(T_5)$ contains a dominating K_4-model and hence contains K_4 as a minor. In particular, $G[V(T_2) \cup \dots \cup V(T_5)]$ contains a subdivision S of K_4 as a subgraph. Let $b_1,\dots,b_4$ be the four vertices of degree 3 in S (the "branch-vertices" of the subdivision). Then by definition of a

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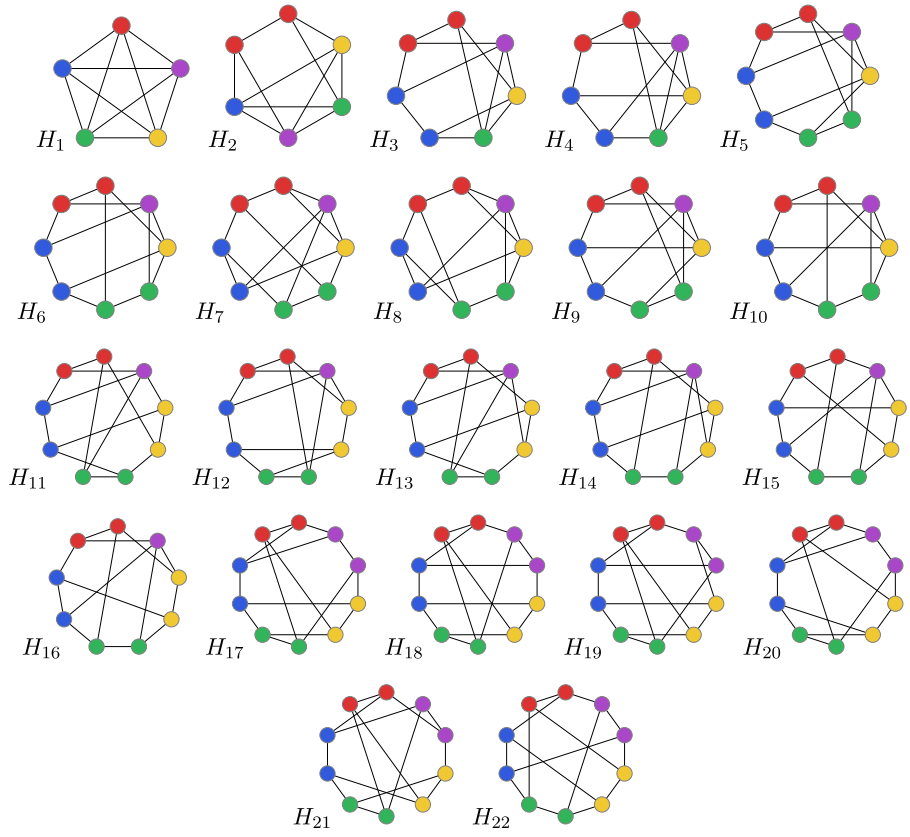


Fig. 1. The 22 graphs obtained by splitting K_5, where vertices of the same colour represent pairs split from the same vertex of K_5.

dominating model, we have that for each $i \in \{1,2,3,4\}$ there exists some $v_i \in V(T_1)$ such that $v_i b_i \in E(G)$ (the v_i are not necessarily distinct). Now consider the smallest subtree T of T_1 containing all of $\{v_1,\dots,v_4\}$. It is easy to see that every leaf of T must lie in $\{v_1,\dots,v_4\}$. It follows that T can be written as the union of five internally vertex-disjoint (but possibly of length 0) paths $P, P_1,\dots,P_4$ such that P_i has one endpoint equal to $v_{\pi(i)}$ for each $i \in \{1,2,3,4\}$ and some permutation $\pi \in S_4$, and such that P_1, P_2 start at one of the endpoints of P while P_3, P_4 start at the other endpoint of P. It can now be easily seen that the graph $S \cup P \cup P_1 \cup P_2 \cup P_3 \cup P_4 \cup \{v_1b_1, v_2b_2, v_3b_3, v_4b_4\}$ forms a subdivision of K_5 if P has length 0, and a subdivision of $\widehat{K}_5$ if P has positive length. Hence, G indeed contains a subdivision of K_5 or $\widehat{K}_5$, concluding the proof. ■

Note that Corollary 1.2 can be strengthened further by requiring that two incident edges of K_5 or two incident edges of $\widehat{K}_5$ in its unique K_4-subgraph are not subdivided (i.e., remain edges) in the subdivision found in G. This is because every graph with a dominating K_4-model contains a subdivision of K_4 with two incident edges not subdivided and so, in the proof of Corollary 1.2, one may assume that S has this property.

Connected Dominating Sets. Further motivation for the dominating Hadwiger conjecture and the study of dominating K_t-models comes from the closely related notion of a *connected dominating set (CDS)*, which is a set of vertices that dominates the graph and induces a connected subgraph. CDS's are

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natural and fundamental objects of study in graph theory, and have many applications in theoretical computer science. In particular, upper bounds on the minimum size of a CDS [1, 4, 5, 17, 29] and the complexity of approximating smallest CDS's [20, 27] have received significant attention. Since they provide easily reachable connected subgraphs, CDS's have proven useful for a variety of algorithmic problems in theoretical computer science such as routing and backbone formation [9, 11] and load balancing [10]; see [27] for an overview of these algorithmic applications. Notably, much research has focused on *packings of disjoint CDS's* in graphs, since such packings are useful to guarantee robust connectivity properties [7, 8]. They have also found applications to seemingly disparate topics such as the design of wireless networks [12, 13], and the study of graph rigidity [30]. We refer the interested reader to the recent paper [16] for new results and a summary of the rich literature on this active topic. To see the connection between packings of disjoint CDS's and dominating K_t-models, note that every packing of t disjoint CDS's in a graph automatically yields a dominating K_t-model. Conversely, for every dominating K_t-model $(T_1, \dots, T_t)$ in a graph, $V(T_i)$ is a CDS of the subgraph induced by $V(T_i) \cup \dots \cup V(T_t)$ for every i. Hence, while not equivalent, a dominating K_t-model can be seen as a weaker version of the notion of a collection of t disjoint CDS's. Similarly to packings of disjoint CDS's, dominating K_t-models provide well-connected subsets of a graph; for example, it can be seen directly from the definition that for every dominating K_t-model $(T_1, \dots, T_t)$, every pair of two distinct vertices in T_i can be joined by i internally vertex-disjoint paths.

2 Proof sketch

One technical difficulty with dominating models is that a minor H of a graph G may contain a dominating K_5-model even if G itself does not (consider the 1-subdivision of K_5). To address this, we use an induction hypothesis that is even stronger than Theorem 1.1. For a vertex v and a dominating K_5-model, $\mathcal{T} = (T_1, \dots, T_5)$, we say that $\mathcal{T}$ is *v-compatible* if $v \notin V(T_2) \cup \dots \cup V(T_5)$. Crucially, if a graph does not contain a v-compatible dominating K_5-model, and a connected subgraph that contains v is contracted into a new vertex v^*, then the resulting graph does not contain a v^*-compatible dominating K_5-model (Lemma 3.2(a)).

For this sketch we aim to prove the following: for every graph G and every vertex $v \in V(G)$, if G does not contain a v-compatible dominating K_5-model, then G is 4-colourable. The actual proof needs an even stronger induction hypothesis (see Lemma 3.1) but the current one will suffice to give a flavour of the proof while avoiding some technical difficulties.

Let us try to prove this result by induction. Let G be a graph that does not contain a v-compatible dominating K_5-model. If G has a vertex of degree at most 3, then we may delete such a vertex, apply induction, and add the vertex back giving it a colour distinct from its neighbours. Hence, we may assume that G has minimum degree at least 4. If G is not 2-connected, then we may split G into two graphs that meet on at most one vertex, apply induction to each of these, relabel the colours so the colourings agree on the common vertex (if there is one), and combine to give a 4-colouring of G. Now suppose that G is not 3-connected and so we may split G into two graphs, G_1 and G_2, that meet on exactly two vertices. Although induction implies that both sides are 4-colourable, these colourings may not agree on the two common vertices. Nonetheless, taking $v \in V(G_1)$ say, we may carefully contract G_1 into the two common vertices and apply induction to ensure that there is a 4-colouring of G_2 that agrees with the 4-colouring of G_1.

To build a dominating K_5-model in G, we need G to be 4-connected. The current induction hypothesis is not sufficiently strong to guarantee there are 4-colourings on both sides of a cut of size 3 that agree. Hence, to deal with the case where G is 3-connected but not 4-connected, we need the even stronger induction hypothesis of Lemma 3.1. This strengthened induction hypothesis and a rerouting argument inspired by [28] deal with this case. For the purposes of this sketch we will just assume that G is 4-connected.

Now consider the subgraph N of G induced by the neighbourhood of v. If N has minimum degree 3, then it contains a dominating K_4-model (Lemma 3.4) and so G contains a v-compatible dominating K_5-model. Thus, we may assume there is a vertex u that is adjacent to v and has at most two common neighbours with v. Let M be the set of vertices adjacent to u and not to v (this includes v). Suppose that M is an independent set. Contract all edges of the form um ($m \in M$) into a vertex v^*. By induction the resulting graph has a

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4-colouring c. This can be turned into a 4-colouring of G by using colour $c(v^*)$ on M and noting that at most three colours appear on the neighbourhood of u, so there is a free colour for u.

If M is not an independent set, then there is a triangle $v_3v_4v_5$ that does not contain v. We now aim to build a v-compatible dominating K_5-model, which would be a contradiction. This is done in Lemma 3.5. The idea is approximately as follows. The 4-connectivity of G gives a subdivision, T, of K_4 that does not contain v and where one of the triangles is just $v_3v_4v_5$. Note that $(T - \{v_3, v_4, v_5\}, v_3, v_4, v_5)$ is a dominating K_4-model in G that does not contain v. By taking T to be in some sense minimal and applying a lemma of [34, (3.1)], we can ensure that $G - V(T)$ is connected and vertices of $G[V(T)]$ have small degree. Since G has minimum degree at least 4, one would hope that all vertices of T have a neighbour outside of $V(T)$ and so $(G - V(T), T - \{v_3, v_4, v_5\}, v_3, v_4, v_5)$ is the required v-compatible dominating K_5-model. This works for all vertices of T except for v_3, v_4, v_5. In practice, we initially build a slightly more precise structure than T to guarantee that v_3, v_4, v_5 all have neighbours outside of T or G is 4-colourable. This is done by combining (3.4) and (3.5) of [34] and is the one place in the proof where we use the 4-colour theorem.

3 The proof

In this paper all graphs are simple. We frequently consider a minor of a graph that is obtained by contracting a set of edges. After the contraction, all resulting loops are removed and parallel edges are replaced by single edges.

We prove Theorem 1.1 by induction with a strengthened induction hypothesis. For a dominating K_5-model $\mathcal{T} = (T_1, \dots, T_5)$ in a graph G and a vertex $v \in V(G)$, define

$$\mathbf{index}_{\mathcal{T}}(v) \coloneqq \begin{cases} i & \text{if } v \in V(T_i) \text{ for } 1 \leqslant i \leqslant 5; \\ 0 & \text{if } v \in V(G) \setminus V(T_1 \cup \cdots \cup T_5). \end{cases}$$

Consider an ordered clique $L = (v_1, \dots, v_k)$ in a graph G with $k \leqslant 2$. A dominating K_5-model $\mathcal{T}$ is *L-compatible* if $\mathbf{index}_{\mathcal{T}}(v_i) \leqslant i$ for each $i \in \{1, \dots, k\}$, and if $k = 2$ and $\mathbf{index}_{\mathcal{T}}(v_2) = 2$, then $\mathbf{index}_{\mathcal{T}}(v_1) = 1$.

Lemma 3.1. (Main Induction Hypothesis) For every graph G and every ordered clique $L = (v_1, \dots, v_k)$ in G with $k \leqslant 2$, either:

- G is 4-colourable, or
- G has an L-compatible dominating K_5-model.

We introduce some tools before proving Lemma 3.1. The following lemma is used to carry L-compatible dominating K_5-models through the induction steps. If edge contractions are done, the position of the resulting vertex in L is the lowest position of any vertex of L among the identified vertices (and if no vertex of L is among the contracted vertices, then the resulting vertex is not in L).

Lemma 3.2. Let $L = (v_1, \dots, v_k)$ be an ordered clique in a graph G with $k \leqslant 2$.

(a) Let H_1 be a connected subgraph of G that contains v_1. Let (G', L') be the pair obtained from (G, L) after contracting the edges of H_1. If G' has an L'-compatible dominating K_5-model, then G has an L-compatible dominating K_5-model.

(b) Let H_2 be a connected subgraph of $G[N_G(v_1)]$ that contains v_2. Let (G', L') be the pair obtained from (G, L) after contracting the edges of H_2. If G' has an L'-compatible dominating K_5-model, then G has an L-compatible dominating K_5-model.

(c) Let $A \subseteq V(G)$. If $G[A]$ has an $(L \cap A)$-compatible dominating K_5-model, then G has an L-compatible dominating K_5-model.

(d) Let L' be an ordered clique on at most two vertices such that L is an initial segment of L'. If G has an L'-compatible dominating K_5-model, then G has an L-compatible dominating K_5-model.

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Proof. We prove the items in turn. Let $\mathcal{T} = (T_1, \dots, T_5)$ be an L'-compatible dominating K_5-model in $G' = G/E(H_1)$. Let v_1^* be the vertex obtained from the contracted edges. Since $v_1 \in V(H_1)$, v_1^* is the first vertex of L' and so $\mathbf{index}_{\mathcal{T}}(v_1^*) \leqslant 1$. First, suppose that $\mathbf{index}_{\mathcal{T}}(v_1^*) = 0$. If $v_2 \in V(H_1)$, then L is disjoint from $\bigcup_{i=1}^5 T_i$ and so $\mathcal{T}$ is L-compatible in G. If $v_2 \notin V(H_1)$, then, since $\mathbf{index}_{\mathcal{T}}(v_1^*) = 0$ and $\mathcal{T}$ is $L' = (v_1^*, v_2)$-compatible, $\mathbf{index}_{\mathcal{T}}(v_2) \leqslant 1$. Again, $\mathcal{T}$ is L-compatible in G. If $\mathbf{index}_{\mathcal{T}}(v_1^*) = 1$, then $(T_1', T_2, \dots, T_5)$ is an L-compatible dominating K_5-model in G, where T_1' is obtained from T_1 by uncontracting the edges of H_1 regardless of whether $v_2 \in V(H_1)$.

Let $\mathcal{T} = (T_1, \dots, T_5)$ be an L'-compatible dominating K_5-model in $G' = G/E(H_2)$. Let v_2^* be the vertex obtained from the contracted edges. Since $v_2 \in V(H_2)$ and $v_1 \notin V(H_2)$, v_2^* is the second vertex of L' and so $\mathbf{index}_{\mathcal{T}}(v_2^*) \leqslant 2$. If $\mathbf{index}_{\mathcal{T}}(v_2^*) = 0$, then $\mathcal{T}$ is also L-compatible in G. If $\mathbf{index}_{\mathcal{T}}(v_2^*) = 1$, then $(T_1', T_2, \dots, T_5)$ is an L-compatible dominating K_5-model in G, where T_1' is obtained from T_1 by uncontracting the edges of H_2. Finally, if $\mathbf{index}_{\mathcal{T}}(v_2^*) = 2$, then $\mathbf{index}_{\mathcal{T}}(v_1) = 1$ and so $(T_1, T_2', T_3, T_4, T_5)$ is L-compatible where T_2' is obtained from T_2 by uncontracting the edges of H_2. Also, since $V(H_2) \subseteq N_G(v_1)$, it remains a dominating K_5-model.

Let $\mathcal{T} = (T_1, \dots, T_5)$ be an $(L \cap A)$-compatible dominating K_5-model in $G[A]$. If $v_1 \in A$, then $\mathcal{T}$ is an L-compatible dominating K_5-model in G. If $v_1 \notin A$, then $\mathbf{index}_{\mathcal{T}}(v_2) \leqslant 1$ (since v_2 is either the first vertex of $L \cap A$ or is not in $G[A]$) and so $\mathcal{T}$ is still L-compatible in G.

Let $\mathcal{T} = (T_1, \dots, T_5)$ be an L'-compatible dominating K_5-model in G. Since L is an initial segment of L', $\mathcal{T}$ is also L-compatible. ■

Recall that a *proper separation* (A, B) of a graph G is a pair $A, B \subset V(G)$ where $A \cup B = V(G)$, both $A \setminus B$ and $B \setminus A$ are non-empty, and there is no edge from $A \setminus B$ to $B \setminus A$. Further, (A, B) is a *proper* ($\leqslant t$)*-separation* (*proper t-separation*) if it is a proper separation and $|A \cap B| \leqslant t$ ($|A \cap B| = t$). Recall that if a graph is not $(k+1)$-connected, then either it is a complete graph on at most $k+1$ vertices or it has a proper ($\leqslant k$)-separation.

Here is a simple lemma that stitches together colourings of $G[A]$ and $G[B]$.

Lemma 3.3. Let $c \geqslant 1$. Suppose that (A, B) is a proper ($\leqslant c$)-separation of a graph G. Let G' be obtained from $G[B]$ by adding a clique K on $A \cap B$. Suppose that $G[A]$ is c-colourable and, for every $E' \subseteq E(K)$, G'/E' (G'/E' is the graph obtained from G' by contracting the edges of E') is c-colourable. Then G is c-colourable.

Proof. Let $\chi_A \colon A \to \{1, \dots, c\}$ be a c-colouring of $G[A]$ and define $S_i := \chi_A^{-1}(i) \cap A \cap B$ for $i \in \{1, \dots, c\}$. Let G'' be obtained from G' by contracting each S_i into a single vertex s_i (if $S_i = \varnothing$, then there is no s_i). By supposition, G'' is c-colourable. Note that the s_i form a clique in G''. Let $\chi_B \colon V(G'') \to \{1, \dots, c\}$ be a c-colouring of G'' with $\chi_B(s_i) = i$ if s_i exists. Let χ be χ_A on A and χ_B on $B \setminus A$. Then χ is a c-colouring of G. ■

We need the following technical strengthening of a result by [26, Thm. 7] that finds dominating K_4-models.

Lemma 3.4. Let G be a connected graph with at least four vertices. Let L be a clique in G on at most two vertices, such that every vertex of degree at most 2 in G is in L, and if some vertex in L has degree at least 3 in G, then $|L| = 1$. Then G has a dominating K_4-model (T_1, T_2, T_3, T_4) with $L \subseteq V(T_1)$.

Proof. We proceed by induction on $|V(G)|$. If $|V(G)| = 4$, then $G = K_4$ and $|L| \leqslant 1$, and the claim is immediate. Now assume $|V(G)| \geqslant 5$ and the result holds for graphs with fewer than $|V(G)|$ vertices. Note that, by including an arbitrary vertex of G in L if L is empty, we may assume L is non-empty.

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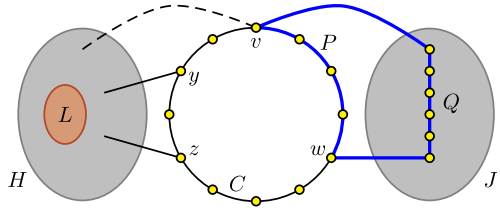


Fig. 2. Proof of Lemma 3.4.

Suppose G is not 2-connected. So G has a proper 1-separation (A, B). Let v be the unique vertex in $A \cap B$. Since L is a clique, without loss of generality, $L \subseteq A$. Note every vertex of $B \setminus A$ has degree at least 3 and so $|B| \geqslant 4$. Apply induction to $G[B]$ with $L' := \{v\}$, which is valid since v is the only possible vertex with degree at most 2 in $G[B]$. So $G[B]$ has a dominating K_4-model (T_1, T_2, T_3, T_4) with $v \in T_1$. Then $(T_1 \cup G[A], T_2, T_3, T_4)$ is a dominating K_4-model in G with $L \subseteq V(T_1) \cup A$. Now assume that G is 2-connected.

Suppose that $G - L$ is a forest. If $G - L$ has an isolated vertex v, then the only neighbours of v are in L, implying $\mathbf{deg}(v) \leqslant |L| \leqslant 2$ and v is in L, which is a contradiction. Thus, $G - L$ contains at least two leaves, denoted v and w. Since v and w are not in L, they both have degree at least 3, implying that v and w both have at least two neighbours in L. So $|L| = 2$, and every vertex in L is adjacent to both v and w. Since L is a clique, every vertex in L has degree at least 3, which contradicts the assumption that $|L| = 1$ in this case.

Hence, $G - L$ is not a forest and so there is a pair (H, C) such that C is a cycle, H is a component of $G - C$, and $L \subseteq V(H)$. Choose such a pair with $|V(H)|$ maximal and, subject to this, $|V(C)|$ minimal. Then C is an induced cycle. Since G is 2-connected, there are distinct vertices y and z in C both adjacent to H.

Suppose there is a vertex $v \in V(C)$ not adjacent to H; refer to Figure 2. So $v \notin \{y, z\}$. Since $v \notin L$, $\mathbf{deg}(v) \geqslant 3$. Since C is induced, v is adjacent to a component J of $G - C$ distinct from H. Since G is 2-connected, there is a vertex w in $C - v$ adjacent to J. Without loss of generality, $w \neq y$. Let P be the vw-path in C avoiding y. Let Q be a vw-path through J. So $P \cup Q$ is a cycle, and there is a component of $G - (P \cup Q)$ that includes H and y (and so contains L), which contradicts the maximality of $|V(H)|$.

Now assume that every vertex in C is adjacent to H. Let xy be an edge of C. Then $(H, C - x - y, \{x\}, \{y\})$ is a dominating K_4-model with $L \subseteq V(H)$. ■

Note that Lemma 3.4 implies that every graph on at least four vertices that does not contain a dominating K_4-model has two non-adjacent vertices of degree at most 2. The corresponding result for K_4-minor-free graphs is well known (see, e.g., [14, Prop. 7.3.1]).

Our final preliminary lemma deals with one of the cases where G is 4-connected.

Lemma 3.5. Let G be a 4-connected graph and $e = v_1v_2$ an edge of G. If G contains a triangle that is vertex-disjoint from e, then either G is planar or G has a (v_1, v_2)-compatible dominating K_5-model.

Proof. The proof uses the following concept of [34]. A *triad with feet* x, y, z, as displayed in Figure 3, consists of an ax-path, an ay-path, and an az-path such that the only vertex on at least two of the paths is a, where $a \notin \{x, y, z\}$ (in particular, each path has length at least 1 and the degrees of x, y, z in the triad are all 1).

Let x, y, z be three distinct vertices of a graph G. Under the assumption that G is 4-connected, (3.5) of [34] implies that either G is planar or has a "tripod" on x, y, z. We will not need the precise definition of a tripod since, under the assumption that G is 4-connected, (3.4) of [34] implies that if G has a tripod on x, y, z, then it has a "legless tripod" on x, y, z: a pair of edge-disjoint triads T_1, T_2, both with feet x, y, z, such that $V(T_1) \cap V(T_2) = \{x, y, z\}$. Combining these two statements we obtain that if x, y, z are distinct vertices

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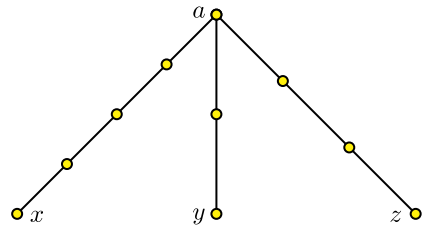


Fig. 3. A triad.

of a 4-connected graph G, then either G is planar or contains a pair of edge-disjoint triads T_1, T_2, both with feet x, y, z, such that $V(T_1) \cap V(T_2) = \{x, y, z\}$.

We now begin the main part of the proof. Let $v_3v_4v_5$ be a triangle that is vertex-disjoint from e. Since G is 4-connected, the previous paragraph implies that either G is planar or there are two edge-disjoint triads T_1, T_2, both with feet v_3, v_4, v_5, such that $V(T_1) \cap V(T_2) = \{v_3, v_4, v_5\}$. Since G is 4-connected, $G - \{v_3, v_4, v_5\}$ is connected and so without loss of generality there is a path from v_1 to T_1 that is vertex-disjoint from T_2 (this includes the case where $v_1 \in V(T_1)$). In particular, there is a component W of $G - V(T_2)$ such that $\{v_1\} \cup (T_1 - \{v_3, v_4, v_5\}) \subseteq W$.

A triad T is *lean* if, for every triad T' with the same feet as T, $V(T')$ is not a proper subset of $V(T)$. Under the assumption that G is 4-connected, (3.1) of [34] says that if T_0 is a triad with feet x, y, z and C is a connected component of $G - V(T_0)$ then there is a lean triad T with feet x, y, z such that T and C are vertex-disjoint and $G - V(T)$ is connected. Applying this here to T_2 and W gives a lean triad T with feet v_3, v_4, v_5 such that $G - V(T)$ is connected and contains W. We claim that $\mathcal{T} = (G - V(T), T - \{v_3, v_4, v_5\}, v_3, v_4, v_5)$ is a (v_1, v_2)-compatible dominating K_5-model.

First $v_1 \in W \subseteq G - V(T)$ and so $\mathbf{index}_{\mathcal{T}}(v_1) = 1$. Since $v_3v_4v_5$ is disjoint from e, $\mathbf{index}_{\mathcal{T}}(v_2) \leqslant 2$ and so $\mathcal{T}$ is (v_1, v_2)-compatible. Second, every part of $\mathcal{T}$ is connected. Third, $v_3v_4v_5$ is a triangle and so the dominating condition between the third, fourth, and fifth parts is satisfied. Next, fix $i \in \{3, 4, 5\}$. Vertex v_i has a neighbour in each of $T - \{v_3, v_4, v_5\}$ and $T_1 - \{v_3, v_4, v_5\}$. Since $T_1 - \{v_3, v_4, v_5\} \subseteq W \subseteq G - V(T)$ the dominating condition holds between the third/fourth/fifth parts and the first/second parts of $\mathcal{T}$. It remains to check that every vertex of $T - \{v_3, v_4, v_5\}$ has a neighbour outside T. Since G is 4-connected, G has minimum degree at least 4, so it suffices to show that every vertex of $T - \{v_3, v_4, v_5\}$ has at most three neighbours in $V(T)$.

Suppose, towards a contradiction, that there is a vertex x of $T - \{v_3, v_4, v_5\}$ with at least four neighbours in T. Let a be the common vertex of the three paths that make up T and let P_i be the av_i-path of T (for $i = 3, 4, 5$). Since T is lean, each P_i is induced in G. In particular, a only has three neighbours in T and so $x \neq a$. Thus, x is an internal vertex of some P_i which we may and will assume is P_3. Since P_3 is induced, x has only two neighbours on P_3. If x has a neighbour in each of $P_4 - a$ and $P_5 - a$, then there is a triad T' with feet v_3, v_4, v_5 where $V(T') \subseteq V(T) \setminus \{a\}$ (see Figure 4(a) where x is the common vertex of the three paths). This contradicts the leanness of T.

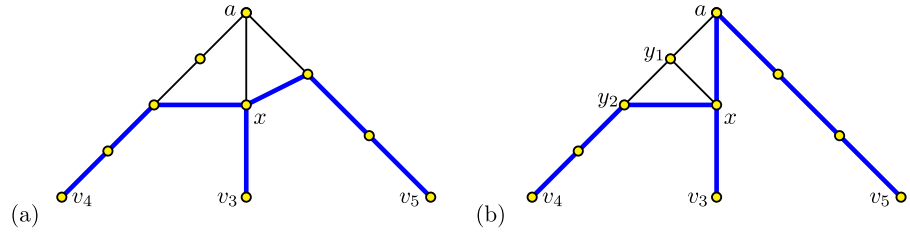


Fig. 4. Triads in the proof of Lemma 3.5.

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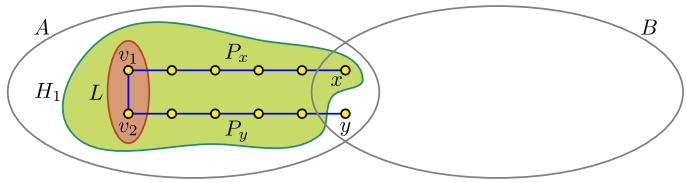


Fig. 5. Eliminating a 2-separation.

Thus, we may and will assume that x has at least two neighbours on $P_4 - a$. In particular, there are vertices y_1, y_2 on $P_4 - a$ both adjacent to x such that y_1 lies between a and y_2 on P_4. But then there is a triad T' with feet v_3, v_4, v_5 where $V(T') \subseteq V(T) \setminus \{y_1\}$ (see Figure 4(b) where x is the common vertex of the three paths). This contradicts the leanness of T, as required. ■

We are now ready to prove Lemma 3.1, which implies Theorem 1.1.

Proof of Lemma 3.1. Suppose, for a contradiction, that there is a pair (G, L) where G is a non-4-colourable graph, L is an ordered clique in G of size at most 2, and G has no L-compatible dominating K_5-model. Choose such a pair (G, L) where $|V(G)|$ is minimal and, subject to this, $|L|$ is maximal. Since G is not 4-colourable, $|V(G)| \geqslant 5$.

Claim. Every vertex of G has degree at least 4.

Proof of Claim. Suppose that there is a vertex $x \in V(G)$ with $\deg(x) \leqslant 3$. By the minimality of $|V(G)|$, either $G - x$ is 4-colourable or $G - x$ has an $L \setminus \{x\}$-compatible dominating K_5-model. In the former case, G is 4-colourable and in the latter case, Lemma 3.2(c) implies that G has an L-compatible dominating K_5-model, which is the required contradiction.

Claim. G is 2-connected.

Proof of Claim. If not, then, since $|V(G)| \geqslant 5$, there is a proper ($\leqslant 1$)-separation (A, B) of G. Since G does not have an L-compatible dominating K_5-model, Lemma 3.2(c) implies that $G[A]$ does not have an $(L \cap A)$-compatible dominating K_5-model. Minimality of $|V(G)|$ implies that $G[A]$ is 4-colourable. Similarly $G[B]$ is 4-colourable. Since $|A \cap B| \leqslant 1$, Lemma 3.3 implies that G is also 4-colourable, which is the required contradiction.

By maximality of $|L|$ and Lemma 3.2(d) we have $|L| = 2$. Let v_1, v_2 be the vertices of L (in that order).

Claim. G is 3-connected.

Proof of Claim. If not, then, since $|V(G)| \geqslant 5$, there is a proper 2-separation (A, B) of G. Since L is a clique, we may and will assume that $L \subseteq A$. Let $\{x, y\} = A \cap B$. Since G does not have an L-compatible dominating K_5-model, Lemma 3.2(c) implies that $G[A]$ does not, and so $G[A]$ is 4-colourable by the minimality of $|V(G)|$.

Without loss of generality, $v_1 \neq y$ and $v_2 \neq x$. As illustrated in Figure 5, since G is 2-connected, y does not separate v_1 from $B \setminus A$ and so there is a (possibly empty) path P_x from v_1 to x in $G[A]$ that avoids y. Similarly there is a path P_y from v_2 to y in $G[A]$. The walk $v_1 v_2 \cup P_y$ shows that there is a path from v_1 to y in $G[A]$. Thus, there is a connected subgraph H_1 of $G[A]$ that contains v_1 and x, does not contain y, but is adjacent to y. Contract $E(H_1)$ to get a vertex v_1^* adjacent to y. Since x was in H_1, we view v_1^* as being in $A \cap B$.

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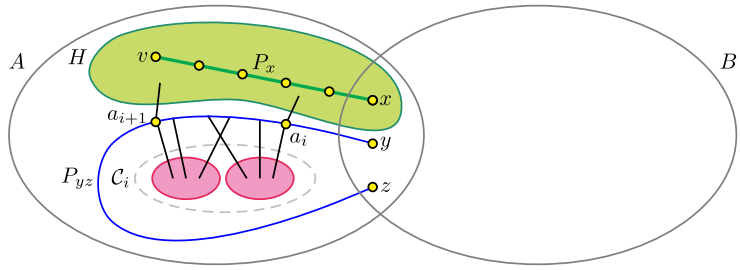


Fig. 6. Eliminating a 3-separation.

The resulting graph, when induced on B, is isomorphic to $G' := G[B] + xy$. Let L' be what L has now become intersected with B. Since $v_1 \in V(H_1)$, the first vertex of L' is v_1^*. Further, in G', contract the edge v_1^*y. Let L'' be what L' has now become. The resulting graph, G'', is isomorphic to $(G[B] + xy)/xy$.

Since G has no L-compatible dominating K_5-model and the contracted subgraphs were connected and contained v_1, Lemma 3.2(a) and (c) imply that G' does not have an L'-compatible dominating K_5-model and G'' does not have an L''-compatible dominating K_5-model. The minimality of $|V(G)|$ implies that G' and G'' are both 4-colourable.

Recall that $G[A]$ is 4-colourable. Thus, Lemma 3.3 implies that G is 4-colourable, which is the required contradiction.

Claim. G is 4-connected.

Proof of Claim. If not, then, since $|V(G)| \geqslant 5$, there is a proper 3-separation (A, B) of G with $L \subseteq A$. Choose such a separation with $|A|$ minimal. Let $\{x, y, z\} = A \cap B$. Since G does not have an L-compatible dominating K_5-model, Lemma 3.2(c) implies that neither does $G[A]$, and so $G[A]$ is 4-colourable by the minimality of $|V(G)|$.

Since every vertex of G has degree at least 4, we have $|A \setminus B| \geqslant 2$. Let $v \in A \setminus B$ be distinct from v_1. Duplicate the vertex v to get a non-adjacent twin v'. Doing this does not change connectivity and so the resulting graph is 3-connected. By Menger's theorem, there are three vertex-disjoint paths between $\{v_1, v, v'\}$ and $\{x, y, z\}$. Without loss of generality, these are P_x from v_1 to x, P from v to y, and P' from v' to z. All three paths must have all their vertices in $A \cup \{v'\}$, since $\{x, y, z\}$ separates v_1, v, v' from $B \setminus A$. In particular, P_x is a v_1x-path in $G[A]$, and in the original graph G, the paths P and P' correspond to paths in $G[A]$ from v to y and v to z that are vertex-disjoint from P_x. Thus, there is a walk from y to z in $G[A]$ that is vertex-disjoint from P_x. In conclusion, $G[A]$ contains the v_1x-path P_x and some yz-path that is vertex-disjoint from P_x.

We now run a standard rerouting argument (see, e.g., [28, Lemma 2.1]). As illustrated in Figure 6, let P_{yz} be a yz-path in $G[A]$ that is vertex-disjoint from P_x and such that:

(i) the number of vertices in the same component as P_x in $G[A] - V(P_{yz})$ is maximised and, subject to this,
(ii) $|V(P_{yz})|$ is minimised.

Replacing P_{yz} by a strictly shorter path from y to z contained in $V(P_{yz})$ would not decrease the number of vertices counted in (i) and strictly decrease the number of vertices counted in (ii), a contradiction. Hence, P_{yz} must be an induced path. Let H be the component of $G[A] - V(P_{yz})$ that contains P_x. We now show that H is the only component of $G[A] - V(P_{yz})$. Let $y = a_0, a_1, \ldots, a_r = z$ be distinct vertices of P_{yz} appearing in that order on P_{yz} such that $\{a_1, \ldots, a_{r-1}\}$ is exactly the set of internal vertices of P_{yz} that have a neighbour in H.

Consider some component C of $G[A] - V(P_{yz})$ that is not H (there may be no such component). Further, suppose that C has neighbours b and c on P_{yz} such that there is some a_i strictly between b and c. Then

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rerouting $bP_{yz}c$ (we use $bP_{yz}c$ to denote the subpath of P_{yz} with ends b and c) through C increases the size of H (it now contains a_i). This contradicts (i). Thus, for every component $C \neq H$ of $G[A]-V(P_{yz})$, there is some $i \in \{0,\dots,r-1\}$ such that all neighbours of C on P_{yz} are in $a_iP_{yz}a_{i+1}$.

For each $i \in \{0,\dots,r-1\}$, let $\mathcal{C}_i$ be the union of the components of $G[A]-V(P_{yz})$ that are not equal to H and whose neighbours on P_{yz} are all in $a_iP_{yz}a_{i+1}$. Every component of $G[A]-V(P_{yz})$, other than H, is contained in some $\mathcal{C}_i$. Now, the neighbourhood, in $G[A]$, of a vertex of $\mathcal{C}_i$ is contained in $\mathcal{C}_i$ and $a_iP_{yz}a_{i+1}$. This is also true for vertices of P_{yz} strictly between a_i and a_{i+1} (recall that P_{yz} is induced and $\{a_1,\dots,a_{r-1}\}$ is exactly the set of internal vertices of P_{yz} that have a neighbour in H). Thus, in $G[A]-\{a_i,a_{i+1}\}$, the component containing any vertex of $\mathcal{C}_i$ is contained in the union of $\mathcal{C}_i$ and the subpath of P_{yz} strictly between a_i and a_{i+1}. Suppose some $\mathcal{C}_i$ is non-empty. Then $\{a_i,a_{i+1}\}$ separates $\mathcal{C}_i$ from $\{x,y,z\}$ in $G[A]$, which implies that $\{a_i,a_{i+1}\}$ separates $\mathcal{C}_i$ from $\{x,y,z\}$ in G. This contradicts the 3-connectivity of G. Therefore, every $\mathcal{C}_i$ is empty and so H is indeed the only component of $G[A]-V(P_{yz})$.

We now show that every vertex of P_{yz} has a neighbour in H. If a is an internal vertex of P_{yz}, then it has degree at least 4 in G and is not in $\{x,y,z\}$, so it has degree at least 4 in $G[A]$. Since P_{yz} is an induced path, a has a neighbour in $G[A]$ that is not on P_{yz} and so is in H. Next consider y and suppose towards a contradiction that y has no neighbours in H. Since P_{yz} is induced, y has only one neighbour in P_{yz} and so only one neighbour in $G[A]$, which we denote by y'. Vertex y is neither equal nor adjacent to $v_1 \in H$, so $y \notin L$. But now $(A\setminus\{y\}, B\cup\{y'\})$ is a proper ($\leqslant 3$)-separation of G (the intersection is $\{x,y',z\}$) with $L \subseteq A\setminus\{y\}$. This contradicts either the minimality of $|A|$ or that G is 3-connected. Therefore y has a neighbour in H. Repeating the same argument for z, we find that also z, and hence indeed every vertex of the path P_{yz}, must have some neighbour in H.

We now contract some edges of $G[A]$ to create a clique on $A\cap B$. First, in the graph G contract all of H into a single vertex v_1^*. Let G' and L' be what G and L have become. Since G does not have an L-compatible dominating K_5-model and $v_1 \in V(H)$, Lemma 3.2(a) implies that G' does not have an L'-compatible dominating K_5-model. Since $v_1 \in V(H)$, v_1^* is the first vertex of L'. If $|L'|=2$, then $v_2 \notin V(H)$ and so, since $L \subseteq A$ and H is the unique component of $G[A]-V(P_{yz})$, vertex $v_2 \in V(P_{yz})$. If $|L'|=1$, then add y as the second vertex of L'. By Lemma 3.2(d), G' still does not have an L'-compatible dominating K_5-model. Furthermore, the first vertex of L' is v_1^* and the second vertex is in $V(P_{yz})$. There is a subpath, P', of P_{yz} that contains the second vertex of L' and all the edges of P_{yz} except for either the first one or the last one. Since every vertex of P_{yz} had a neighbour in H, we have $V(P') \subset N_{G'}(v_1^*)$. Contract P' to a single vertex so y and z are now adjacent. Let G'' and L'' be what G' and L' have become. Lemma 3.2(b) implies that G'' does not have an L''-compatible dominating K_5-model. Note that $G'' \cong G[B]+E(K)$ where K is a clique on $\{x,y,z\}$.

Let $E'' \subseteq E(K)$. Let $G''' := G''/E''$ and L''' be what G'' and L'' become after contracting the edges of E''. If $E''=\varnothing$, then $G'''=G''$ and $L'''=L''$. Then G''' does not have an L'''-compatible dominating K_5-model and so is 4-colourable by the minimality of $|V(G)|$. Otherwise, since K is a triangle, the edges of E'' form a connected graph (which does not necessarily span $V(K)$). If that subgraph contains v_1^*, then Lemma 3.2(a) implies that G''' does not contain an L'''-compatible dominating K_5-model and so is 4-colourable by the minimality of $|V(G)|$. Otherwise $E''=\{yz\}$. Recall that we ensured the second vertex of L' was in $V(P_{yz})$. Thus, the second vertex of L'' is one of y or z. Hence, Lemma 3.2(b) implies that G''' does not contain an L'''-compatible dominating K_5-model and so is 4-colourable by the minimality of $|V(G)|$.

We have shown that, for every $E'' \subseteq E(K)$, $G'' \cong G[B]+E(K)$ satisfies $\chi(G''/E'') \leqslant 4$. Recall that $G[A]$ is 4-colourable. Lemma 3.3 implies that G is 4-colourable which is the required contradiction.

Now, if G contains a triangle that is vertex-disjoint from L, then, by Lemma 3.5, either G is planar (and is thus 4-colourable; this is the one place in the proof where we use the 4-colour theorem) or G has an L-compatible dominating K_5-model, a contradiction. Thus, $G-L$ is triangle-free.

Let $N := G[N_G(v_1)]$. Note that $v_2 \in V(N)$ and N has at least four vertices so there exists $x \in V(N)\setminus\{v_2\}$. Suppose that every vertex of $N-\{v_2\}$ has degree at least 3 in N. Let C be the component of N containing x. Then C is connected and has at least four vertices. Let $L' := V(C)\cap\{v_2\}$ so L' is a clique on at most one

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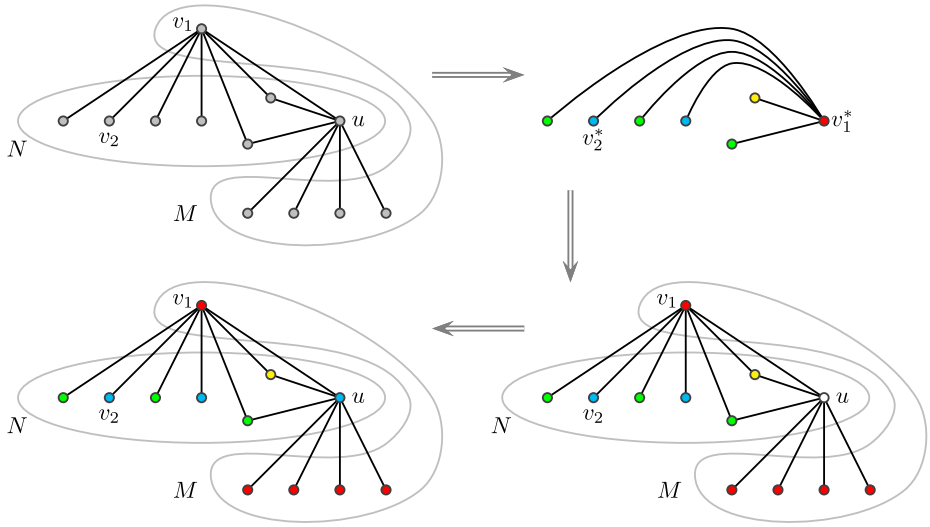


Fig. 7. Colouring steps in the 4-connected case.

vertex. Also, every vertex of degree at most 2 in C is in L'. Lemma 3.4 then implies that C has a dominating K_4-model (T_2, T_3, T_4, T_5) with $L' \subseteq V(T_2)$. In particular, $v_2 \notin T_3 \cup T_4 \cup T_5$. But then $(\{v_1\}, T_2, T_3, T_4, T_5)$ is an L-compatible dominating K_5-model in G, a contradiction.

Thus, there is some vertex $u \in V(N) \setminus \{v_2\}$ with at most two neighbours in N. As illustrated in Figure 7, let M be the set of neighbours of u that are not adjacent to v_1 (this includes v_1). Since $G - L$ is triangle-free, $M \setminus \{v_1\}$ is an independent set. Since M is disjoint from $N_G(v_1)$, M is an independent set. Let G' be obtained from G by contracting the edges of $\{um\colon m \in M\}$ (these span a connected subgraph of G that contains v_1) into a new vertex v_1^*. Let L' be what L has now become. By Lemma 3.2(a), G' does not have an L'-compatible dominating K_5-model, so G' has a 4-colouring χ by the minimality of $|V(G)|$. Colour G as follows. Use colouring χ on $G - (\{u\} \cup M)$ and colour the vertices of M with colour $\chi(v_1^*)$. It remains to colour u. Since $|N_G(u) \setminus M| = \deg_N(u) \leqslant 2$, at most three colours appear on the neighbourhood of u in G and so there is a colour available for u. This gives a 4-colouring of G. This contradiction completes the proof. ■

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